Rockset Workload Isolation Architecture

Zero Data Latency, Virtual Instances & Compute-Storage Decoupling

< 1 Second	100% Isolated
GUARANTEED DATA FRESHNESS FROM STREAMING KAFKA	COMPUTE BOUNDARIES BETWEEN INGESTION, APPS & BI

1. The Real-Time Dual Constraint

In operational analytics, systems must deliver two opposing guarantees:

- **Zero Data Latency:** Mutated business events (orders, telemetry, clicks) must be queryable via SQL in < 1 second without batch ETL pipeline lag.
- **Strict Workload Isolation:** Large streaming ingestion spikes or heavy BI queries must never compromise the sub-50ms p99 SLA of customer-facing applications.

2. Why Monolithic Shared-Resource Databases Fail

When writes and queries share CPU cores, memory buffers, and disk IOPS, performance collapses under concurrent load:

Shared Contention Vector	Physical Failure Mechanism	Impact on Latency SLA
Buffer Pool Contention	Heavy writes flush dirty pages to memory, evicting query cache blocks.	Subsequent read queries stall on cold disk seeks.
CPU Thread Starvation	Complex analytical joins monopolize all available CPU cores.	Ingestion threads stall; upstream Kafka logs lag.
Replication Lag on Replicas	Offloading queries to standard read replicas introduces asynchronous lag.	Violates freshness; users view stale, inconsistent data.

3. The Virtual Instance (VI) Solution

Rockset decouples compute and storage using **RocksDB-Cloud**. Storage durability is anchored in shared cloud object storage (AWS S3), while independent, dedicated clusters called **Virtual Instances (VIs)** execute distinct workloads with dedicated CPU, RAM, and NVMe cache allocations.

4. How Zero Data Latency (< 1s) Operates Across Isolated VIs

Because Query VIs and Ingest VIs run on completely separate servers, synchronizing mutations without duplicating storage requires a specialized metadata protocol:

The Micro-Batch Metadata Synchronization Protocol

1. Continuous MemTable Appends: Ingest VI appends records to local RocksDB MemTables and WAL.
2. Micro-Batch SST Flushes: Every 500ms, Ingest VI flushes immutable SSTables and uploads them to S3.
3. Lightweight Commit Manifest Broadcast: Ingest VI publishes a tiny metadata manifest (S3 paths, DocID ranges) over an in-memory pub-sub bus (< 50ms latency).
4. Atomic Version Advance: Query VIs receive the manifest and update their local RocksDB `VersionSet`. Data is immediately queryable via S3 zero-copy reads!

5. Latency Formulation from Kafka to Query Visibility

The end-to-end data freshness latency budget is mathematically bounded by:

$$T_{freshness} = T_{ingest_pull} + T_{memtable_flush} + T_{manifest_sync} \approx 20ms + 500ms + 50ms \approx 570\ ms$$

6. Eliminating LSM Compaction Interference

In vanilla RocksDB, background SST compactions consume 40% of disk and CPU bandwidth, causing "write stalls" and query latency spikes.

$$Workload_Query \cap Workload_Compaction = \varnothing$$

Rockset delegates all compactions to an autoscaling fleet of **Serverless Remote Compactors**. Neither the Ingest VI nor the Query VIs run compactions locally, ensuring query tail latency remains strictly unaffected by background database maintenance.

7. Multi-Workload Topology in Production

Virtual Instance Role	Compute Profile	SLA Latency Target	Failure Domain Isolation
Ingest VI	Memory & Network Heavy	Continuous streaming (>100k writes/s)	Isolated; unaffected by long-running queries.
Live Customer App VI	Compute & NVMe Optimized	**p99 < 50 ms**	100% isolated; zero latency degradation.
Internal BI / Analytics VI	Elastic High-Core Count	Seconds to minutes (Ad-hoc exploratory)	Can be paused or throttled without user impact.

8. Empirical Isolation Stress-Test Benchmarks

Cluster Operating State	Monolithic Shared Cluster (p99)	Rockset Virtual Instance Cluster (p99)
Baseline Query Load	24 ms	**22 ms**
During 10× Ingest Write Burst	185 ms (7.7× degradation)	**23 ms (Zero degradation)**
Concurrent Heavy BI Queries + Ingest Burst	1,240 ms (51× degradation)	**24 ms (Completely Isolated)**
Observed Data Freshness	18 – 45 seconds (Replication lag)	**< 1 second (Guaranteed)**

9. Architectural Lessons & Production Guardrails

1. Anchor Durability in Object Storage

Using cloud object storage (S3) as the single source of truth allows spinning up isolated compute clusters on demand without replicating raw underlying datasets.

2. Synchronize Versions via Metadata

Broadcasting lightweight commit manifests (< 50ms) ensures sub-second freshness across isolated clusters without copying heavy data payloads across networks.